

Draft - expert review pending

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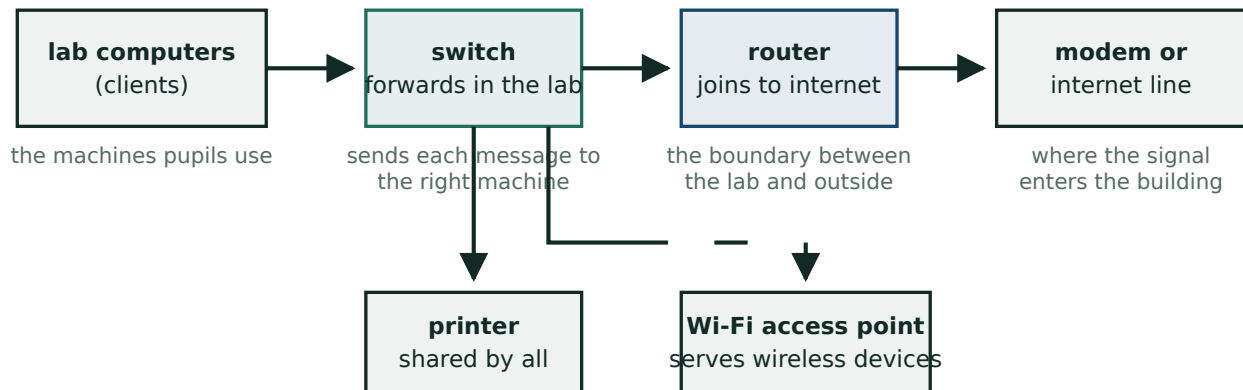
Topic 2.4 - Communication hardware and software

A real classroom situation

The new computer lab in a Sukkur school has twenty machines, one printer and one internet connection. On the first day, the technician connects everything, and the head teacher asks the obvious question: "So all twenty computers can print to the one printer and use the one internet line?" They can, and the reason is a small box on the shelf with blinking lights - plus the invisible rules that let machines take turns. A teacher who can name the parts and the rules can also diagnose the lab's most common complaint: "the internet is not working on computer 14 only."

How a school lab's computers connect

The chain from pupil machines to the internet, and the devices that share it



Every device gets an IP address; shared rules (protocols) let them take turns.

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Explanation

Communication hardware and software

A computer network is a set of computers and devices connected so they can share resources: files, a printer, an internet connection. Bourgeois and colleagues (2019) describe the internet itself as an interconnected network of networks, and a school lab is a small working model of the same idea.

The hardware of a small network divides into four jobs. The network interface (built into every modern computer and phone) is each machine's doorway onto the network. The switch connects multiple machines in one place and forwards each message only to the machine it is addressed to, which is why twenty computers can share one printer without shouting over each other. The router is the device that connects one network to another: it receives messages, analyses them, and routes them towards their destination (Bourgeois et al., 2019). In the lab, the router is the boundary between the school's network and the internet line. The medium carries the signal: a copper cable across the room, fibre for longer distances, or radio for Wi-Fi, which takes an internet signal and converts it into radio waves (Bourgeois et al., 2019).

Every device on a network is assigned a unique identifying number called an IP address, and a human-friendly name - a domain name - is translated to that number by the DNS system, which acts as the directory of the internet (Bourgeois et al., 2019). When the lab's internet "does not work", one of these named parts is usually at fault, and the names are what let you describe the fault to someone who can fix it.

The software side of communication is everything that makes the hardware usable. A protocol is a set of rules that governs how communication takes place on a network (Bourgeois et al., 2019); machines can only talk when both follow the same rules, which is why standards matter so much. On top of the protocols run the tools you actually see: the browser that requests web pages, the email client, the messaging and video-meeting apps, the software that shares a folder or a printer across the lab. The operating system contributes too: it holds the network settings

- the address, the password, the connection name - that let the machine join.

Communication hardware and software at a glance

Item	Hardware or software	What it does
network interface card	hardware	the machine doorway onto the network
switch	hardware	forwards messages to the right machine in the lab
router	hardware	connects the school network to the internet
Wi-Fi access point	hardware	serves wireless devices in range
cable or fibre	hardware	carries the signal
protocol	software rules	the shared rules machines follow to talk
browser and email client	software	request pages and send mail
network settings in the OS	software	hold the address and password the machine joins with

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A note on the everyday Pakistan reality: in many schools the network is not a lab of cables but a phone hotspot and a data bundle, with the school's single

router doubling as the Wi-Fi source for the head teacher's laptop. The parts are the same parts, just fewer of them. Naming them does not depend on scale.

Activity: Map the lab's network

Work in pairs. About 20 minutes.

1. If your classroom has a network (or your school has one), walk it and sketch what you find: how many computers, where the printer is, what the boxes with lights are, what connects to what. If no network is available, sketch the one in the figure from the labels alone.
2. Label each part in your sketch with its job: doorway, forwarder, boundary, medium.
3. Add the software layer: write beside the sketch which tools carry the school's actual communication (which browser, which email, which messaging app, and where the Wi-Fi password lives).
4. Finish with one diagnostic sentence in the form: "If the internet works on computer 5 but not on computer 14, the fault is more likely in ... than in ..., because ...".

Check your understanding

1. Define a computer network in one sentence. (*Remember*)
2. State the difference between a switch and a router in one sentence each. (*Understand*)
3. In the sketch of a lab network, the printer works for every computer but no computer can reach the internet. Name the part most likely at fault and justify your answer. (*Apply*)
4. Name two communication tools that are software and one rule-set that is a protocol. (*Remember*)

Summary

Communication hardware forms a chain: each machine's network interface, the switch that forwards messages inside the lab, the router that joins the lab to the

internet, and the medium (cable, fibre or Wi-Fi radio) that carries the signal. The software side is the shared rules - protocols - plus the visible tools: browsers, email, messaging, and the network settings in the operating system. A teacher who can name these parts can describe any fault precisely, which is half of fixing it.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name the four hardware jobs in a small network and one device for each.
- ☐ I can explain what a protocol is and why machines need shared rules.
- ☐ I can sketch the lab network and say where the internet enters it.

Try this at your practicum school

Sketch the practicum school's actual network (or its single-router setup). Label every part you can identify, and mark the one point where the internet enters the building. Ask the in-charge what usually breaks first.

Summative task

A remote school gets a laptop, a printer, a Wi-Fi router and an internet connection, and the head teacher wants teachers to share the printer and the internet from their own devices. Draw the network (boxes and lines), label each part's job, and write four sentences: two on how the hardware must be arranged, and two on the software settings that must be right for it to work.

Marking guide (10 marks):

- All devices placed and connected correctly in the diagram: 3 marks.
- Each part's job labelled accurately: 3 marks.
- The software sentences name actual settings or tools, not vague "setup" (Analyse): 3 marks.

- Diagram is clear and labelled: 1 mark.

Further reading

- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 5. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Laudon, K. C., & Laudon, J. P. *Management Information Systems*. Pearson. (Course guide reading; print monograph, cited at bibliographic level.)
- Stair, R., & Reynolds, G. *Principles of Information Systems*. Cengage Learning. (Course guide reading; print monograph, cited at bibliographic level.)

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